

Becoming More Fluent with Function Notation and Reviewing Previous Math

Finding an Input of a Function Given an Output

- ① If $f(x) = 2x - 3$ and $f(b) = 9$, find b .

$$f(x) = 2x - 3$$

$$f(b) = 2b - 3$$

$$9 = 2b - 3$$

$$12 = 2b$$

$$\boxed{b = 6}$$

- ③ If $f(x) = 7 - 5x$ and $f(m) = -2$, find m .

$$f(x) = 7 - 5x$$

$$f(m) = 7 - 5m$$

$$-2 = 7 - 5m$$

$$-9 = -5m$$

$$\boxed{m = \frac{9}{5}}$$

- ② If $f(x) = 4 - \frac{x}{9}$ and $f(c) = 10$, find c .

$$f(x) = 4 - \frac{x}{9}$$

$$f(c) = 4 - \frac{c}{9}$$

$$10 = 4 - \frac{c}{9}$$

$$6 = -\frac{c}{9}$$

$$\boxed{c = -54}$$

- ④ If $f(x) = \frac{x}{2}$ and $f(a) = 1$, find a .

$$f(x) = \frac{x}{2}$$

$$f(a) = \frac{a}{2}$$

$$1 = \frac{a}{2}$$

$$\boxed{a = 2}$$

- ⑤ If $f(x) = 4x + 3$ and $f(k) = -8$, find k .

$$f(x) = 4x + 3$$

$$f(k) = 4k + 3$$

$$-8 = 4k + 3$$

$$-11 = 4k$$

$$\boxed{k = -\frac{11}{4}}$$

- ⑦ If $f(x) = -3x + 10$ and $f(r) = 25$, find r .

$$f(x) = -3x + 10$$

$$f(r) = -3r + 10$$

$$25 = -3r + 10$$

$$\boxed{r = -5}$$

- ⑥ If $f(x) = 3 + \frac{x}{5}$ and $f(p) = 4$, find p .

$$f(x) = 3 + \frac{x}{5}$$

$$f(p) = 3 + \frac{p}{5}$$

$$4 = 3 + \frac{p}{5}$$

$$1 = \frac{p}{5}$$

$$\boxed{p = 5}$$

- ⑧ If $f(x) = 6 - \frac{x}{4}$ and $f(v) = -1$, find v .

$$f(x) = 6 - \frac{x}{4}$$

$$f(v) = 6 - \frac{v}{4}$$

$$-1 = 6 - \frac{v}{4}$$

$$-7 = -\frac{v}{4}$$

$$\boxed{28 = v}$$

- ⑨ If $f(x) = x^2 + 9x + 10$, and $f(b) = -4$, find b .

$$f(x) = x^2 + 9x + 10$$

$$f(b) = b^2 + 9b + 10$$

$$-4 = b^2 + 9b + 10$$

$$0 = b^2 + 9b + 14$$

$$0 = (b+2)(b+7)$$

↙ ↘

$$\boxed{b = -2 \text{ or } b = -7}$$

- ⑪ If $f(x) = x^2 + 3x - 1$ and $f(k) = 9$, find k .

$$f(x) = x^2 + 3x - 1$$

$$f(k) = k^2 + 3k - 1$$

$$9 = k^2 + 3k - 1$$

$$0 = k^2 + 3k - 10$$

$$0 = (k+5)(k-2)$$

↙ ↘

$$\boxed{k = -5 \text{ or } k = 2}$$

- ⑩ If $f(x) = x^2 - 3x - 26$, and $f(c) = 2$, find c .

$$f(x) = x^2 - 3x - 26$$

$$f(c) = c^2 - 3c - 26$$

$$2 = c^2 - 3c - 26$$

$$0 = c^2 - 3c - 28$$

$$0 = (c-7)(c+4)$$

↙ ↘

$$\boxed{c = 7 \text{ or } c = -4}$$

- ⑫ If $f(x) = x^2 - 2x - 6$ and $f(m) = -7$, find m .

$$f(x) = x^2 - 2x - 6$$

$$f(m) = m^2 - 2m - 6$$

$$-7 = m^2 - 2m - 6$$

$$0 = m^2 - 2m + 1$$

$$0 = (m-1)(m-1)$$

↙ ↘

$$\boxed{m = 1 \text{ or } m = 1}$$

- ⑬ If $f(x) = 10x^2 + 17x$ and $f(d) = -3$, find d .

$$f(x) = 10x^2 + 17x$$

$$f(d) = 10d^2 + 17d$$

$$-3 = 10d^2 + 17d$$

$$0 = 10d^2 + 17d + 3$$

$$0 = (2d+3)(5d+1)$$

↙ ↘

$$\boxed{d = -\frac{3}{2} \text{ or } d = -\frac{1}{5}}$$

- ⑮ If $f(x) = -9x^2 + x + 1$ and $f(k) = 1$, find k .

$$f(x) = -9x^2 + x + 1$$

$$f(k) = -9k^2 + k + 1$$

$$1 = -9k^2 + k + 1$$

$$0 = -9k^2 + k$$

$$0 = k(-9k+1)$$

$$\boxed{k = 0 \text{ or } k = \frac{1}{9}}$$

- ⑭ If $f(x) = 7x^2 - 43x - 30$ and $f(r) = -36$, find r .

$$f(x) = 7x^2 - 43x - 30$$

$$f(r) = 7r^2 - 43r - 30$$

$$-36 = 7r^2 - 43r - 30$$

$$0 = 7r^2 - 43r + 6$$

$$0 = (7r-1)(r-6)$$

↙ ↘

$$\boxed{r = \frac{1}{7} \text{ or } r = 6}$$

- ⑯ If $f(x) = 2x^2 - 4x - 4$ and $f(b) = -4$, find b .

$$f(x) = 2x^2 - 4x - 4$$

$$f(b) = 2b^2 - 4b - 4$$

$$-4 = 2b^2 - 4b - 4$$

$$0 = 2b^2 - 4b$$

$$0 = 2b(b-2)$$

↙ ↘

$$\boxed{b = 0 \text{ or } b = 2}$$

- (17) If $f(x) = 4x^2 - 5x - 3$ and $f(m) = 4$, find m .

$$f(x) = 4x^2 - 5x - 3$$

$$f(m) = 4m^2 - 5m - 3$$

$$4 = 4m^2 - 5m - 3$$

$$0 = 4m^2 - 5m - 7$$

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$m = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(4)(-7)}}{2(4)}$$

$$m = \frac{5 \pm \sqrt{25 + 112}}{8}$$

$$m = \frac{5 \pm \sqrt{137}}{8}$$

- (18) If $f(x) = -2x^2 + 7x - 3$ and $f(p) = -8$, find p .

$$f(x) = -2x^2 + 7x - 3$$

$$f(p) = -2p^2 + 7p - 3$$

$$-8 = -2p^2 + 7p - 3$$

$$0 = -2p^2 + 7p + 5$$

$$p = \frac{-7 \pm \sqrt{7^2 - 4(-2)(5)}}{2(-2)}$$

$$p = \frac{-7 \pm \sqrt{49 + 40}}{-4}$$

$$p = \frac{-7 \pm \sqrt{89}}{-4}$$

$$p = \frac{7 \pm \sqrt{89}}{4}$$

- (19) If $f(x) = 4 + \frac{(\frac{x}{3} - 1)^2}{2}$ and $f(c) = 6$, find c .

$$f(x) = 4 + \frac{(\frac{x}{3} - 1)^2}{2}$$

$$f(c) = 4 + \frac{(\frac{c}{3} - 1)^2}{2}$$

$$6 = 4 + \frac{(\frac{c}{3} - 1)^2}{2}$$

$$2 = \frac{(\frac{c}{3} - 1)^2}{2}$$

$$4 = (\frac{c}{3} - 1)^2$$

$$\pm \sqrt{4} = \sqrt{(\frac{c}{3} - 1)^2}$$

$$\pm 2 = \frac{c}{3} - 1$$

$$2 = \frac{c}{3} - 1 \text{ or } -2 = \frac{c}{3} - 1$$

$$c = 9 \text{ or } c = -3$$

②0 If $f(x) = -3(11x+4)^2 - 1$ and $f(v) = -6$, find v .

$$f(x) = -3(11x+4)^2 - 1$$

$$f(v) = -3(11v+4)^2 - 1$$

$$-6 = -3(11v+4)^2 - 1$$

$$-5 = -3(11v+4)^2$$

$$\frac{5}{3} = (11v+4)^2$$

$$\pm \sqrt{\frac{5}{3}} = 11v+4$$

$$\pm \frac{\sqrt{5}}{\sqrt{3}} = 11v+4$$

$$\pm \frac{\sqrt{5} \cdot \sqrt{3}}{\sqrt{3} \cdot \sqrt{3}} = 11v+4$$

$$\pm \frac{\sqrt{15}}{\sqrt{9}} = 11v+4$$

$$\pm \frac{\sqrt{15}}{3} = 11v+4$$

$$\pm \frac{\sqrt{15}}{3} = 11v+4$$

$$\pm \frac{\sqrt{15}}{3} - 4 = 11v$$

$$\frac{\pm \frac{\sqrt{15}}{3} - 4}{11} = v$$

$$\frac{3\left(\pm \frac{\sqrt{15}}{3} - 4\right)}{3(11)} = v$$

$$\boxed{\frac{\pm \sqrt{15} - 12}{33} = v}$$

• ②1 If $f(x) = \frac{(-5x+4)^2}{3} - 8$ and $f(b) = -5$, find b .

$$f(x) = \frac{(-5x+4)^2}{3} - 8$$

$$f(b) = \frac{(-5b+4)^2}{3} - 8$$

$$-5 = \frac{(-5b+4)^2}{3} - 8$$

$$3 = \frac{(-5b+4)^2}{3}$$

$$9 = (-5b+4)^2$$

$$\pm 3 = -5b+4$$

$$\swarrow \quad \searrow$$

$$3 = -5b+4 \text{ or } -3 = -5b+4$$

$$\boxed{b = \frac{1}{5} \text{ or } b = \frac{7}{5}}$$

②② If $f(x) = 2\left(\frac{x}{3} + 6\right)^2 + 10$ and $f(r) = 15$, find r .

$$f(x) = 2\left(\frac{x}{3} + 6\right)^2 + 10$$

$$f(r) = 2\left(\frac{r}{3} + 6\right)^2 + 10$$

$$15 = 2\left(\frac{r}{3} + 6\right)^2 + 10$$

$$5 = 2\left(\frac{r}{3} + 6\right)^2$$

$$\frac{5}{2} = \left(\frac{r}{3} + 6\right)^2$$

$$\pm \sqrt{\frac{5}{2}} = \frac{r}{3} + 6$$

$$\pm \frac{\sqrt{5}}{\sqrt{2}} = \frac{r}{3} + 6$$

$$\pm \frac{\sqrt{5} \cdot \sqrt{2}}{\sqrt{2} \cdot \sqrt{2}} = \frac{r}{3} + 6$$

$$\pm \frac{\sqrt{10}}{\sqrt{4}} = \frac{r}{3} + 6$$

$$\pm \frac{\sqrt{10}}{2} = \frac{r}{3} + 6$$

$$\pm \frac{\sqrt{10}}{2} - 6 = \frac{r}{3}$$

$$\boxed{\pm \frac{3\sqrt{10}}{2} - 18 = r}$$

• ②③ If $f(x) = 4\sqrt{3x+1} + 9$ and $f(b) = 13$, find b .

$$f(x) = 4\sqrt{3x+1} + 9$$

$$f(b) = 4\sqrt{3b+1} + 9$$

$$13 = 4\sqrt{3b+1} + 9$$

$$4 = 4\sqrt{3b+1}$$

$$1 = \sqrt{3b+1}$$

$$(1)^2 = (\sqrt{3b+1})^2$$

$$1 = 3b+1$$

$$0 = 3b$$

$$\boxed{b=0}$$

②④ If $f(x) = -3\sqrt{\frac{x}{2}-7} + 5$ and $f(c) = -6$, find c .

$$f(x) = -3\sqrt{\frac{x}{2}-7} + 5$$

$$f(c) = -3\sqrt{\frac{c}{2}-7} + 5$$

$$-6 = -3\sqrt{\frac{c}{2}-7} + 5$$

$$-11 = -3\sqrt{\frac{c}{2}-7}$$

$$\frac{11}{3} = \sqrt{\frac{c}{2}-7}$$

$$\left(\frac{11}{3}\right)^2 = \left(\sqrt{\frac{c}{2}-7}\right)^2$$

$$\frac{121}{9} = \frac{c}{2} - 7$$

$$\frac{184}{9} = \frac{c}{2}$$

$$\boxed{\frac{368}{9} = c}$$

- (25) If $f(x) = \frac{\sqrt{3+2x}}{-5} + 1$ and $f(k) = 1$, find k .

$$f(x) = \frac{\sqrt{3+2x}}{-5} + 1$$

$$f(k) = \frac{\sqrt{3+2k}}{-5} + 1$$

$$1 = \frac{\sqrt{3+2k}}{-5} + 1$$

$$0 = \frac{\sqrt{3+2k}}{-5}$$

$$0 = \sqrt{3+2k}$$

$$0 = 3+2k$$

$$\boxed{k = -\frac{3}{2}}$$

- (26) If $f(x) = 2\sqrt{\frac{x}{5}-4} - 10$ and $f(b) = -7$, find b .

$$f(x) = 2\sqrt{\frac{x}{5}-4} - 10$$

$$f(b) = 2\sqrt{\frac{b}{5}-4} - 10$$

$$-7 = 2\sqrt{\frac{b}{5}-4} - 10$$

$$3 = 2\sqrt{\frac{b}{5}-4}$$

$$\frac{3}{2} = \sqrt{\frac{b}{5}-4}$$

$$\frac{9}{4} = \frac{b}{5} - 4$$

$$\frac{25}{4} = \frac{b}{5}$$

$$\boxed{\frac{125}{4} = b}$$

- (27) If $f(x) = \frac{1}{x-3}$ and $f(b) = 2$, find b .

$$f(x) = \frac{1}{x-3}$$

$$f(b) = \frac{1}{b-3}$$

$$2 = \frac{1}{b-3}$$

$$(b-3)2 = \frac{1}{b-3}(b-3)$$

$$2b-6 = 1$$

$$\boxed{b = \frac{7}{2}}$$

- (28) If $f(x) = \frac{2}{4x+5}$ and $f(c) = 7$, find c .

$$f(x) = \frac{2}{4x+5}$$

$$f(c) = \frac{2}{4c+5}$$

$$7 = \frac{2}{4c+5}$$

$$7(4c+5) = 2$$

$$28c+35 = 2$$

$$28c = -33$$

$$\boxed{c = -\frac{33}{28}}$$

- (29) If $f(x) = \frac{-1}{x+2}$ and $f(m) = -10$, find m .

$$f(x) = \frac{-1}{x+2}$$

$$f(m) = \frac{-1}{m+2}$$

$$-10 = \frac{-1}{m+2}$$

$$-10m - 20 = -1$$

$$\boxed{m = -\frac{19}{10}}$$

- (30) If $f(x) = \frac{7}{8-5x}$ and $f(n) = 2$, find n .

$$f(x) = \frac{7}{8-5x}$$

$$f(n) = \frac{7}{8-5n}$$

$$2 = \frac{7}{8-5n}$$

$$16 - 10n = 7$$

$$\boxed{n = \frac{9}{10}}$$

Finding Inputs of Functions That Make Outputs Equal

- (31) If $f(x) = 3x+7$ & $g(x) = -x+1$, for what value of x does $f(x) = g(x)$? (32) If $f(x) = 2x+1$ & $h(x) = 3x+4$, for what value of x does $f(x) = h(x)$?

$$f(x) = g(x)$$

$$3x+7 = -x+1$$

$$4x = -6$$

$$\boxed{x = -\frac{3}{2}}$$

$$f(x) = h(x)$$

$$2x+1 = 3x+4$$

$$\boxed{-3 = x}$$

- (33) If $f(x) = -\frac{1}{2}x-8$ & $g(x) = 2(x-4)$, find the coordinates of the point where the graphs of these functions intersect.

$$\text{I) } f(x) = g(x)$$

$$-\frac{1}{2}x-8 = 2(x-4)$$

$$-\frac{1}{2}x-8 = 2x-8$$

$$0 = \frac{5}{2}x$$

$$0 = x$$

$$\text{II) } f(0) = -\frac{1}{2}(0)-8 = -8$$

$$\boxed{(0, -8)}$$

③④ If $f(x) = 4(x+3) - 7$ & $h(x) = \frac{x}{5} + 2$, find the coordinates of the point where the graphs of these functions intersect.

I) $f(x) = h(x)$

$$4(x+3) - 7 = \frac{x}{5} + 2$$

$$4x + 12 - 7 = \frac{x}{5} + 2$$

$$4x + 5 = \frac{x}{5} + 2$$

$$\frac{19x}{5} = -3$$

$$19x = -15$$

$$x = -\frac{15}{19}$$

II) $f(-\frac{15}{19}) = 4(-\frac{15}{19} + 3) - 7$

$$= 4(\frac{42}{19}) - 7$$

$$= \frac{168}{19} - 7$$

$$= \frac{35}{19}$$

$$\left(-\frac{15}{19}, \frac{35}{19}\right)$$

• ③⑤ If $f(x) = x + 1$ & $g(x) = 2x$, for what value of x does $f(x) = g(x)$?

$$f(x) = g(x)$$

$$x + 1 = 2x$$

$$\boxed{1 = x}$$

③⑥ If $f(x) = 9 - 3x$ & $h(x) = -4(x+1)$, for what value of x does $f(x) = h(x)$?

$$f(x) = h(x)$$

$$9 - 3x = -4(x+1)$$

$$9 - 3x = -4x - 4$$

$$\boxed{x = -13}$$

③⑦ If $f(x) = -10x$ & $g(x) = x + \frac{2}{5}$, find the coordinates of the point where the graphs of these functions intersect.

I) $f(x) = g(x)$

$$-10x = x + \frac{2}{5}$$

$$-11x = \frac{2}{5}$$

$$\left(-\frac{1}{11}\right)(-11x) = \frac{2}{5}\left(-\frac{1}{11}\right)$$

$$x = -\frac{2}{55}$$

II) $f(-\frac{2}{55}) = -10(-\frac{2}{55})$

$$= \frac{20}{55}$$

$$= \frac{4}{11}$$

$$\left(-\frac{2}{55}, \frac{4}{11}\right)$$

③⑧ If $f(x) = \frac{x}{6} - 3$ & $h(x) = 5(x-2) + 10$, find the coordinates of the point where the graphs of these functions intersect.

I) $f(x) = h(x)$

$$\frac{x}{6} - 3 = 5(x-2) + 10$$

$$\frac{x}{6} - 3 = 5x - 10 + 10$$

$$\frac{x}{6} - 3 = 5x$$

$$-3 = \frac{29x}{6}$$

$$-18 = 29x$$

$$x = -\frac{18}{29}$$

II) $f(-\frac{18}{29}) = \frac{-18}{6} - 3$

$$= -\frac{18}{29} \div 6 - 3$$

$$= \left(-\frac{18}{29}\right) \frac{1}{6} - 3$$

$$= -\frac{18}{174} - 3$$

$$= -3\frac{9}{87}$$

$$\boxed{\left(-\frac{18}{29}, -3\frac{9}{87}\right)}$$

• ③⑨ If $f(x) = x^2 + 10x - 9$ & $g(x) = 2x^2 - x + 1$, for what value(s) of x does $f(x) = g(x)$?

$$f(x) = g(x)$$

$$x^2 + 10x - 9 = 2x^2 - x + 1$$

$$0 = x^2 - 11x + 10$$

$$0 = (x-10)(x-1)$$

$$\boxed{x=10 \text{ or } x=1}$$

• ④⑩ If $f(x) = -2x^2 - 4x - 5$ & $h(x) = -3x^2 + x + 9$, for what value(s) of x does $f(x) = h(x)$?

$$f(x) = h(x)$$

$$-2x^2 - 4x - 5 = -3x^2 + x + 9$$

$$x^2 - 5x - 14 = 0$$

$$(x-7)(x+2) = 0$$

$$\boxed{x=7 \text{ or } x=-2}$$

• ④⑪ If $f(x) = 3x^2 + 8x$ & $g(x) = -x^2 + 5$, find the coordinates of the point(s) where the graphs of these functions intersect.

I) $f(x) = g(x)$

$$3x^2 + 8x = -x^2 + 5$$

$$4x^2 + 8x - 5 = 0$$

$$(2x-1)(2x+5) = 0$$

$$x = \frac{1}{2} \text{ or } x = -\frac{5}{2}$$

II) $g(\frac{1}{2}) = -(\frac{1}{2})^2 + 5 = -\frac{1}{4} + 5 = 4\frac{3}{4}$

$$\boxed{\left(\frac{1}{2}, 4\frac{3}{4}\right)}$$

$$g\left(-\frac{5}{2}\right) = -\left(-\frac{5}{2}\right)^2 + 5 = -\frac{25}{4} + 5 = -\frac{5}{4}$$

$$\boxed{\left(-\frac{5}{2}, -\frac{5}{4}\right)}$$

- ④② If $f(x) = 2x^2$ & $h(x) = -3x + 2$, find the coordinates of the point(s) where the graphs of these functions intersect.

I) $f(x) = h(x)$

$$2x^2 = -3x + 2$$

$$2x^2 + 3x - 2 = 0$$

$$(2x-1)(x+2) = 0$$

$$x = \frac{1}{2} \text{ or } x = -2$$

II) $f(\frac{1}{2}) = 2(\frac{1}{2})^2 = 2(\frac{1}{4})$

$$= \frac{1}{2}$$

$$\left(\frac{1}{2}, \frac{1}{2}\right)$$

$$f(-2) = 2(-2)^2 = 2(4)$$

$$= 8$$

$$(-2, 8)$$

- ④③ If $f(x) = -2(3x+1)^2 - 3$ & $h(x) = -9$, for what value(s) of x does $f(x) = h(x)$?

$$f(x) = h(x)$$

$$-2(3x+1)^2 - 3 = -9$$

$$-2(3x+1)^2 = -6$$

$$(3x+1)^2 = 3$$

$$3x+1 = \pm\sqrt{3}$$

$$3x = \pm\sqrt{3} - 1$$

$$x = \frac{\pm\sqrt{3} - 1}{3}$$

- ④④ If $g(x) = \frac{(x-2)^2}{4} + 2$ & $h(x) = 5$, for what value(s) of x does $h(x) = g(x)$?

$$h(x) = g(x)$$

$$5 = \frac{(x-2)^2}{4} + 2$$

$$3 = \frac{(x-2)^2}{4}$$

$$12 = (x-2)^2$$

$$\pm\sqrt{12} = x-2$$

$$\pm\sqrt{4 \cdot 3} = x-2$$

$$\pm\sqrt{4} \cdot \sqrt{3} = x-2$$

$$\pm 2\sqrt{3} = x-2$$

$$\pm 2\sqrt{3} + 2 = x$$

- ④⑤ If $g(x) = 2 + \frac{\sqrt{\frac{x}{2} + 2}}{3}$ & $f(x) = 4$, find the coordinates of the point where the graphs of these functions intersect.

I) $g(x) = f(x)$

$$2 + \frac{\sqrt{\frac{x}{2} + 2}}{3} = 4$$

$$\frac{\sqrt{\frac{x}{2} + 2}}{3} = 2$$

$$\sqrt{\frac{x}{2} + 2} = 6$$

$$\frac{x}{2} + 2 = 36$$

$$\frac{x}{2} = 34$$

$$x = 68$$

II) $f(68) = 4$

$$(68, 4)$$

- ④⑥ If $h(x) = \sqrt{-5x + 2} - 7$ & $f(x) = -6$, find the coordinates of the point where the graphs of these functions intersect.

I) $h(x) = f(x)$

$$\sqrt{-5x + 2} - 7 = -6$$

$$\sqrt{-5x + 2} = 1$$

$$-5x + 2 = 1$$

$$-5x = -1$$

$$x = \frac{1}{5}$$

II) $f\left(\frac{1}{5}\right) = -6$

$$\left(\frac{1}{5}, -6\right)$$